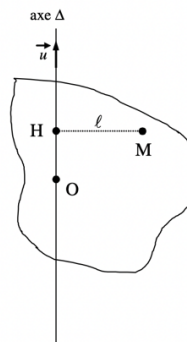
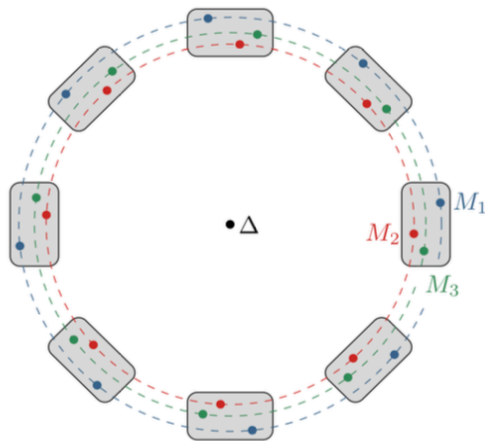


Ch 20
Solide en rotation autour d'un axe fixe

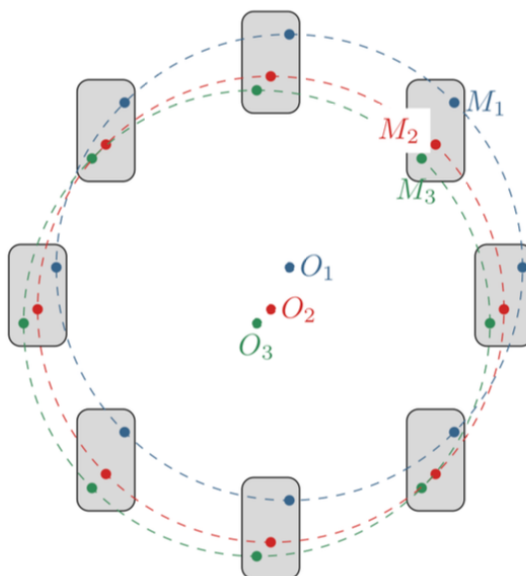
Cinématique



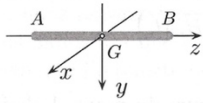
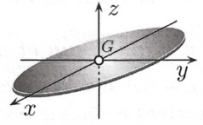
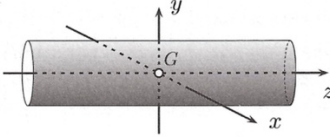
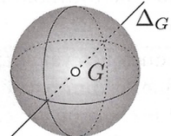
Rotation d'un solide autour d'un axe fixe



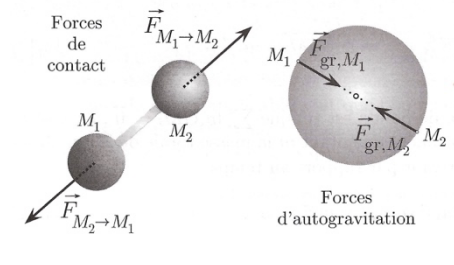
Translation circulaire (grande roue)



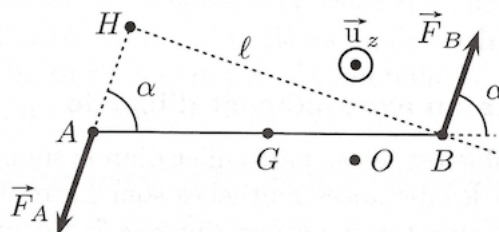
Calcul de moments d'inertie (géométries courantes):

Tige unidimensionnelle homogène de masse m et de longueur ℓ 	$J_{(G, \vec{u}_x)} = J_{(G, \vec{u}_y)} = \frac{1}{12}m\ell^2$ $J_{(B, \vec{u}_x)} = J_{(B, \vec{u}_y)} = \frac{1}{3}m\ell^2$
Disque plein homogène de masse m et de rayon R 	$J_{(G, \vec{u}_z)} = \frac{1}{2}mR^2$ $J_{(G, \vec{u}_x)} = J_{(G, \vec{u}_y)} = \frac{1}{4}mR^2$
Cylindre plein homogène de masse m, de longueur ℓ et de rayon R 	$J_{(G, \vec{u}_z)} = \frac{1}{2}mR^2$ $J_{(G, \vec{u}_x)} = J_{(G, \vec{u}_y)} = \frac{1}{4}mR^2 + \frac{1}{12}m\ell^2$
Sphère pleine homogène de masse m et de rayon R 	$J_{\Delta_G} = \frac{2}{5}mR^2$

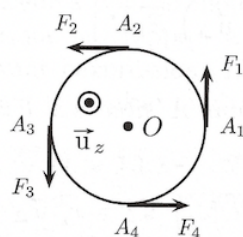
Exemples de forces intérieures à un solide :



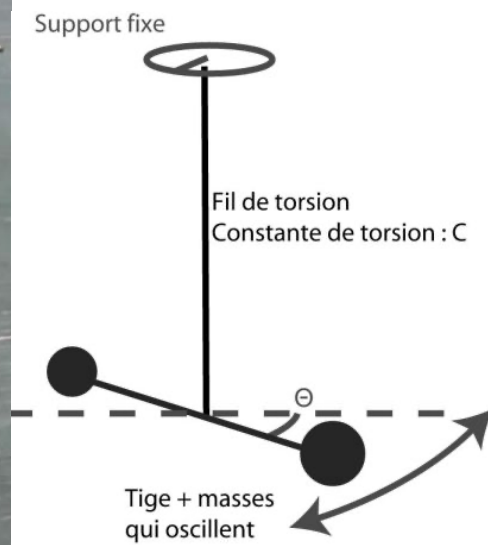
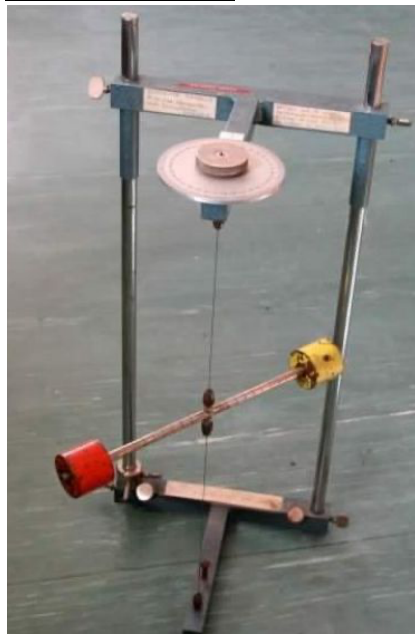
Couple de 2 forces :



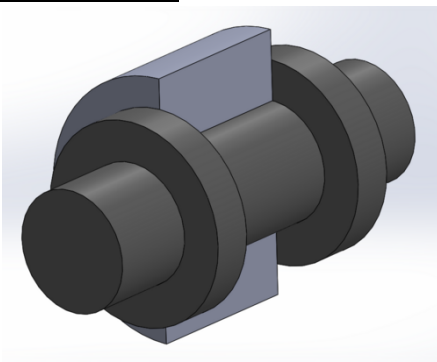
Couple de N forces :



Couple de torsion



Liaison pivot :



Pendule pesant

